

Formulas for Derivatives

1. $\frac{d}{dx}(x^n) = nx^{n-1}$
2. $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$ (Chain rule)
3. $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$
4. $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}$
5. $\frac{d}{dx}(\sin ax) = a \cos ax$
6. $\frac{d}{dx}(\cos ax) = -a \sin ax$
7. $\frac{d}{dx}e^x = e^x$
8. $\frac{d}{dx}e^{u(x)} = u'(x)e^{u(x)}$
9. $\frac{d}{dx}(b^x) = b^x \ln b$
10. $\frac{d}{dx}(b^{u(x)}) = b^{u(x)}(\ln b)u'(x)$
11. $\frac{d}{dx} \ln x = \frac{1}{x}$
12. $\frac{d}{dx}(\log_b x) = \frac{1}{x \ln b}$, for $x > 0$

Trigonometric Substitution Identities

1. $a^2 - x^2 \rightarrow a^2 \cos^2 \theta$ where $x = a \sin \theta$
2. $a^2 + x^2 \rightarrow a^2 \sec^2 \theta$ where $x = a \sin \theta$
3. $x^2 - a^2 \rightarrow a^2 \tan^2 \theta$ where $x = a \sec \theta$

Pythagorean Identity

$$4. \sin^2 \theta + \cos^2 \theta = 1$$

Half-Angle Formulas

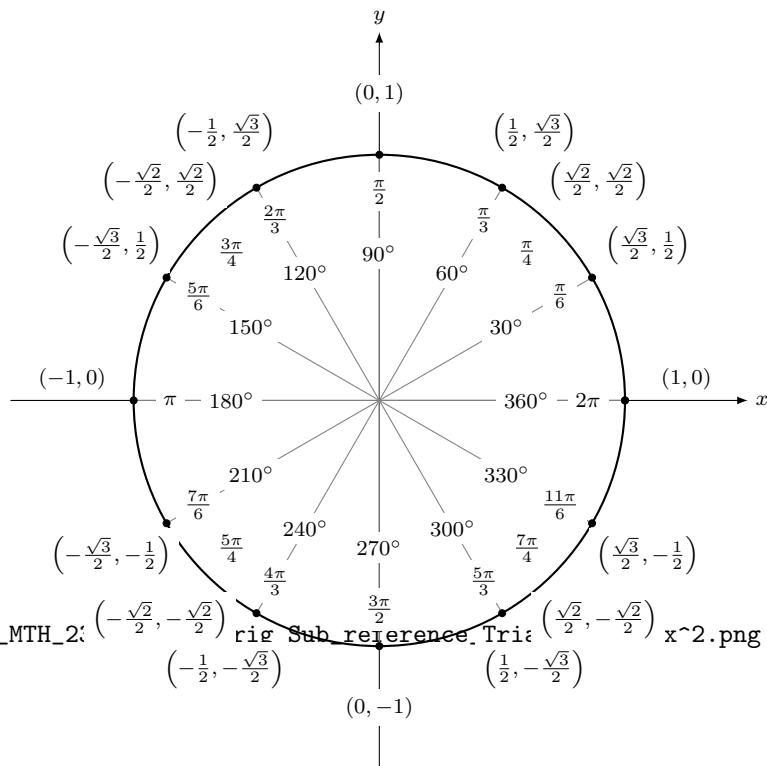
$$5. \cos^2 \theta = \frac{1 + \cos 2\theta}{2}; \quad \sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

Double-Angle Identities

$$6. \sin 2\theta = 2 \sin \theta \cos \theta; \quad \cos 2\theta = 2 \cos^2 \theta - 1$$

Formulas for Integration

1. $\int x^n dx = \frac{1}{n+1}x^{n+1} + C; n \neq -1$
2. $\int \cos ax dx = \frac{1}{a} \sin ax + C$
3. $\int \sin ax dx = -\frac{1}{a} \cos ax + C$
4. $\int e^{ax} dx = \frac{1}{a}e^{ax} + C$
5. $\int \frac{1}{x} dx = \ln |x| + C$
6. $\int b^x dx = \frac{1}{\ln b}b^x + C$ for $b > 0$ and $b \neq 1$
7. $\int \ln x dx = x \ln x - x + C$
8. $\int \tan x dx = \ln |\cos x| + C$
9. $\int \sec x dx = \ln |\sec x + \tan x| + C$
10. $\int f(g(x))g'(x) dx = \int f(u) du$ (Substitution rule)
11. $\int u dv = uv - \int v du$ (Integration by parts)



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